ScholAR: Deploying a Local, Auditable Scientific Document Assistant on Consumer Hardware

Anonymous ACL submission

Abstract

Scientific-document assistants must connect generated statements to inspectable source regions while respecting document privacy and modest hardware budgets. We present ScholAR, a local-first assistant whose ingestion, retrieval, generation, citation mapping, claim verification, and selective repair stages share a versioned execution trace. The system separates acquisition-enabled operation from strict-local analysis, assigns source-scoped identities to evidence, distinguishes model-emitted from application-added citations, and applies deterministic keep, narrow, remap, delete, or abstain actions followed by verification. We also introduce a release protocol that freezes the expected case universe before generation, retains errors and abstentions in denominators, and reconstructs tables from immutable raw traces. This artifact-gated manuscript reports no human-study or model-effectiveness result until independent labels, a paper-disjoint test set, ethics review, a researcher pilot, and final hardware runs are complete. The resulting design exposes a practical lesson for deployed retrieval-augmented systems: provenance, support assessment, and failure policy are separate contracts and must be evaluated separately.

1 Introduction

Retrieval-augmented generation (RAG) can make a language model's evidence available at inference time (Lewis et al., 2020), but a deployed document assistant needs stronger operational contracts. A citation can resolve to a real page and still fail to support the attached statement; an automated support score can be reproducible and still be poorly calibrated; and "local" inference can coexist with network-dependent document or model acquisition. These distinctions matter for scientific papers, where readers routinely inspect methods, equations, tables, figures, and cross-document references.

ScholAR is an expected-use scientific-document assistant for researchers working with local, licensed, unpublished, or otherwise sensitive PDFs. It runs inference through loopback model services on consumer hardware and exposes the evidence associated with each answer. Our deployment question is not whether local RAG is universally more accurate. It is: *which software and evaluation contracts are needed to make local document assistance inspectable, selectively fallible, and reproducible?*

This paper makes three implementation contributions. First, ScholAR uses one canonical ingestion transaction and source-scoped evidence identities so that an answer can be traced to the exact document artifact and page. Second, production and evaluation share a versioned answer pipeline that records retrieved evidence, prompt evidence, generation metadata, citation origin, verification outcomes, text edits, abstentions, timing, hardware tier, and code identity. Third, a deterministic selective-repair policy preserves supported text, narrows partially supported clauses, attempts one evidence remap, removes unsupported or contradicted claims, and abstains when no supported claim remains.

The paper also contributes a fail-closed release method for the eventual deployment study. Expected $(\mathit{system}, \mathit{model}, \mathit{seed}, \mathit{case})$ keys are frozen before generation; success, abstention, and error are immutable terminal outcomes; scoring never invokes generation; and aggregation is case-balanced. Human validation and user evidence remain explicit gates rather than inferred outcomes. This separation is deliberate: software correctness is necessary evidence for deployment, but it is not a substitute for effectiveness, calibration, or human-subject evidence.

2 System and Trust Boundaries

2.1 Canonical document state

ScholAR accepts a local PDF upload or an explicitly acquired document. Publication is transactional: all processing occurs in an isolated sibling staging directory (`.staging-uuid`). Ingestion derives nine canonical artifacts validated by `PaperFinalizeService`:

1. `paper.pdf`: verified source PDF byte stream;
2. `evidence_ast.json`: structural EvidenceAST;
3. `pages.json`: contiguous 1-based page records;
4. `chunks.json`: semantic chunks with section tags;
5. `figures.json`: isolated figure and table metadata;
6. `visual_units.json`: page and crop image descriptors;
7. `metadata.json`: bibliographic metadata and counts;
8. `document.db`: relational SQLite index;
9. `ingestion_manifest.json`: SHA-256 chunk and bundle hashes.

Before publication, the service validates document identity across all files, contiguous page sequences, globally unique block and chunk identifiers, chunk-to-AST referential consistency, figure image file checksums, and SQLite relational constraints. A file-level Unix lock (`fcntl.flock`) serializes concurrent updates. Publication executes via an atomic directory rename (`os.replace`); if replacement fails, the previous valid generation is automatically restored from backup.

2.2 Acquisition-enabled and strict-local modes

The network boundary distinguishes two operational modes. *Acquisition-enabled* mode permits outbound HTTPS requests to paper repositories, package registries, and model repositories. *Strict-local* mode enforces total isolation: `NetworkPolicyService` intercepts all outbound requests prior to DNS resolution or socket creation, rejecting external network access. Only local filesystem caches, prepared uploads, and loopback HTTP endpoints (`127.0.0.1`, `::1`, `localhost`) for local Ollama runtimes are permitted.

2.3 Retrieval, cross-modal expansion, and fusion

Queries undergo intent-driven semantic decomposition, splitting compound questions across clause boundaries into orthogonal sub-intents to detect implicit visual or tabular demands without requiring explicit figure or table keywords.

Retrieval executes across five complementary channels:

- **Lexical:** Okapi BM25 (Robertson and Zaragoza, 2009) with parameters $k_1 = 1.4$ and $b = 0.72$ over camelCase-tokenized chunk texts (with average chunk length $\bar{L}$):

$$\mathrm{BM25}(q, d) = \sum_{t \in q} \mathrm{IDF}(t) \cdot \frac{f(t, d)(k_1 + 1)}{f(t, d) + k_1(1 - b + b|d|/\bar{L})} \quad (1)$$

- **Dense text:** Cosine similarity over dense sentence embeddings (`all-MiniLM-L6-v2`).
- **Crop visual:** Cosine similarity over CLIP embeddings (`openai/clip-vit-base-patch32`) with a strict similarity floor ≥ 0.18.
- **Page visual:** Full-page late-interaction scoring via ColQwen2 (`vidore/colqwen2-v1.0-hf`) using the MaxSim token-level operator:

$$S(Q, P) = \sum_{i=1}^{|Q|} \max_{j=1}^{|P|} \mathbf{q}_i^\top \mathbf{p}_j \quad (2)$$

- **Modality scoring:** Rule-based heuristic boosts (+3.0 for bridged figures cited in top text chunks, +2.5 for comparative result or ablation tables, +2.0 for intent-matched visuals, +1.0 for preferred pages).

Channel ranks are merged using Reciprocal Rank Fusion (RRF) with constant $k = 60$:

$$\mathrm{RRF}(d) = \sum_{c \in \mathcal{C}} \frac{1}{60 + \mathrm{rank}_c(d)} \quad (3)$$

When retrieved text chunks cite visual or tabular elements, *active cross-modal graph expansion* automatically resolves and pins those visual AST nodes into the prompt context. To prevent spurious visual hallucinations, visual hits receive a reranking prior only if corroborated by lexical overlap, bridged citation in text, or explicit figure references. Uncorroborated visual hits are retained strictly as bounded inspection candidates.

2.4 Structured multi-level reasoning paths

To ensure structured reasoning fidelity, ScholAR organizes evidence and generation into multi-level reasoning paths following Sun et al. (2025). Rather than emitting unconstrained chains-of-thought, the pipeline maps intermediate reasoning steps across a structured mode taxonomy (PROBLEMUNDERSTANDING, HYPOTHESISFORMATION, DERIVATION, VERIFICATION, CALCULATION, SYNTHESIS), bounding each intermediate subgoal to ≤ 30 words. Text synthesis organizes responses across three structured tiers: (1) Problem Context and Phenomenon Definition, (2) Mechanistic and Mathematical Derivation, and (3) Quantitative Empirical Evidence with sentence-terminal citation attribution.

2.5 One production and evaluation trace

All user-facing chat endpoints and release evaluation harnesses invoke the unified `AnswerPipelineService`. The resulting `AnswerTrace` stores request parameters, seed, model tag, prompt hash, Git commit identity, multi-channel retrieval scores, exact prompt evidence, raw LLM completion, normalized text, citation origins (model-emitted, application-imputed, or remapped), initial and post-repair verification reports, concrete text edits, abstention reasons, and stage-by-stage latency breakdowns. Evaluation observes this identical execution trace, preventing drift between development evaluation and deployed behavior.

3 Claim Support and Selective Repair

3.1 Span-preserving claims

The verifier segments model-generated text into atomic factual claims while retaining zero-based, half-open character offsets $[s_c, e_c)$ into the exact generated text. Citation markers $[E_k]$ are parsed as distinct citation spans $[s_m, e_m)$, decoupling factual assertions from citation pointers. Each atomic claim records its textual span, associated citation IDs, resolved evidence references, first- and second-pass labels, confidence score, and repair action.

3.2 Support and contradiction scoring

The implemented scorer (`LexicalSupportScorer`) is a deterministic, reproducible baseline rather than an uncalibrated black-box language model. Let T_{claim} denote the non-citation tokens of an atomic claim and T_{evidence} denote the tokens of cited evidence chunks. The token overlap ratio is computed as:

$$\text{Overlap}(T_{\text{claim}}, T_{\text{evidence}}) = \frac{|T_{\text{claim}} \cap T_{\text{evidence}}|}{|T_{\text{claim}}|} \tag{4}$$

To detect factual hallucinations, the verifier extracts numerical quantities, filtering out structural identifiers (e.g., "Figure 1", "Table 2", "Equation 3", "page 4"). If the claim contains non-structural numbers N_{claim} disjoint from evidence numbers N_{evidence} ($N_{\text{claim}} \cap N_{\text{evidence}} = \emptyset$) and exhibits high topical lexical overlap (Overlap > 0.40), it is categorized as CONTRADICTED (confidence $= 0.90$). Defining $\text{Ov} \equiv \text{Overlap}(T_{\text{claim}}, T_{\text{evidence}})$, the claim is otherwise classified by versioned thresholds:

$$\text{Label} = \begin{cases} \text{SUPPORTED}, & \text{Ov} \geq 0.50, \\ \text{PARTIAL}, & 0.25 \leq \text{Ov} < 0.50, \\ \text{UNSUPPORTED}, & \text{Ov} < 0.25. \end{cases} \tag{5}$$

3.3 Conservative deterministic repair actions

Rather than invoking unconstrained LLM regenerations that risk introducing secondary hallucinations, ScholAR executes deterministic, span-preserving text mutations from a bounded action vocabulary:

- `NONE`: SUPPORTED claims are preserved without modification.
- `CLAIM_NARROWING`: For PARTIAL multi-clause claims, the verifier splits clauses and deletes unsupported sub-clauses, retaining only lexical spans supported by cited evidence.
- `CITATION_REMAP`: For an UNSUPPORTED claim, the engine attempts one remap to unused candidate evidence in the retrieved pool; if a candidate meets the 0.50 threshold, citation markers are updated.
- `NUMERIC_CORRECTION` / `CLAIM_DELETION`: Contradicted or stubbornly unsupported claims are excised entirely by deleting their span $[s_c, e_c)$.
- `ABSTAIN`: If repair excises all factual claims, the entire answer is deterministically replaced by the canonical abstention string: *"The paper does not provide sufficient evidence to answer this question."*

Following repair, the text undergoes a second full verification pass. The trace records initial and final reports, exact character diffs, and verification metrics, ensuring complete operational auditability.

4 Evaluation and Release Protocol

4.1 Predeclared evidence gates

The final evaluation protocol enforces strict separation between software infrastructure and empirical effectiveness claims. It mandates a paper-disjoint evaluation set of previously unseen scientific documents with verified answerability ground-truth, supporting text spans, visual regions, and reference key points. Claim-support thresholds must be pre-calibrated on development documents before scoring held-out test splits.

The primary intervention endpoint is the independently judged supported atomic-claim rate. We predeclare a success gate requiring at least five absolute percentage points improvement with a paired 95% bootstrap confidence interval excluding zero, while answer key-point coverage may decrease by at most five percentage points. If this gate fails, the manuscript reports a null or adverse result and cannot claim effective selective repair. Secondary metrics include contradiction rate, citation precision and coverage, answerable versus unanswerable abstention, character edit distance, stage latencies, and terminal error counts.

4.2 Immutable expected-key accounting

Before evaluation begins, the runner defines and serializes the complete Cartesian key universe:

$$\mathcal{K} = \mathcal{S} \times \mathcal{M} \times \mathcal{R} \times \mathcal{C} \quad (6)$$

where $\mathcal{S}$ is the set of system variants, $\mathcal{M}$ is the set of model tags and SHA-256 digests, $\mathcal{R}$ is the set of random seeds, and $\mathcal{C}$ is the evaluation case set.

Every key in $\mathcal{K}$ maps to exactly one JSONL record recording an immutable terminal status: SUCCESS, ABSTAINED, or ERROR. Interrupted runs resume by skipping already-recorded keys; duplicate or missing keys trigger immediate validation failure. Crucially, failures and abstentions remain in all metric denominators. Metrics are aggregated hierarchically: scores are computed within each answer, averaged across seeds within each case, and finally averaged across cases, preventing answers with high claim counts from distorting the aggregate.

4.3 Artifact-only reproduction

The codebase provides a deterministic toy fixture containing multi-seed runs, out-of-order execution traces, and all three terminal statuses. In a strictly isolated, network-denied sandbox, the reproduction runner recomputes scored records, aggregate JSON summaries, and LaTeX tables from raw traces, validating them byte-for-byte against golden checksums. This verifies the integrity of the release tooling independently of empirical model weights.

4.4 External benchmark grounding with SPIQA

To evaluate cross-modal reasoning without artificial prompt hints, ScholAR implements an evaluation adapter for SPIQA (Banerjee et al., 2024). SPIQA benchmarks multimodal question answering over research papers across five canonical visual categories: Architectural Diagrams, Performance Plots, Comparative Tables, Loss Curves, and System Schemas. The adapter evaluates retrieval accuracy (Hit@K, MRR), answer fidelity (token-level macro-F1 and exact match), and visual modality selection precision under real scientific reasoning workloads.

5 Results

Evidence gate pending. The claim-bearing release does not yet exist. Development claim labels, frozen held-out cases, judge validation, ethics determination, researcher pilot, final model-backed intervention runs, and named-hardware profiling remain pending. No historical, simulated, toy, or development-only number is substituted here.

The implemented artifact gates already enforce four reporting properties. First, a failed generation cannot disappear from the denominator. Second, an abstention remains distinct from an infrastructure or model error. Third, tables carry the hash of the aggregate JSON from which they were rendered. Fourth, the submission claim map cannot mark an empirical claim supported without a validated release source. These are reproducibility properties of the evaluation system, not findings about answer quality or user benefit.

The final version of this section will report the paired repair comparison, support–coverage curve, human–judge agreement, user-task result, resource envelope, and failure taxonomy only after their gates clear. If a gate does not clear, the corresponding claim and table will be removed or reported as a null result rather than estimated from development artifacts.

6 Deployment Lessons

Provenance is not support. Stable evidence identifiers and page mappings eliminate fabricated citation locations, but they do not establish entailment. In deployed scientific systems, page validity, page support, claim support, citation precision, and citation coverage represent orthogonal quantities and must be measured independently.

Failure must be a first-class output. Silent retries or substituting fallback text during evaluation hides real operational degradation. ScholAR treats errors and abstentions as immutable outcomes, categorizing infrastructure unavailabilities, evidence insufficiencies, verifier-driven abstentions, and generation errors as distinct terminal states in both telemetry and user interfaces.

Locality is a lifecycle property. Running local LLM inference does not make document ingestion, dependency resolution, or model asset downloads offline. Enforcing an explicit boundary between acquisition-enabled setup and strict-local analysis ensures the trust boundary is auditable and prevents unintended network leaks.

Repair changes the product objective. Deleting unsupported claims maximizes precision but carries a coverage cost. A production evaluation must therefore assess retained coverage, answerable abstention rates, end-to-end latency, and user task completion alongside raw claim support scores. Deterministic repair renders these trade-offs auditable.

Consumer hardware constraints. ScholAR is targeted at consumer-tier hardware, specifically Apple Silicon workstations (16GB–64GB unified memory) and consumer GPUs (16GB–24GB VRAM). Deployments require balancing VRAM allocation between 4-bit quantized local generators (e.g., `qwen2.5:7b-instruct-q4_K_M`), sentence transformers, and multi-vector ColQwen2 indices. Profiling isolates latencies across ingestion, five-channel retrieval, local generation, and verification passes. Malformed PDFs, scanned documents, reading-order ambiguities, and interrupted transactions remain monitored failure modes.

7 Conclusion

ScholAR makes local scientific-document assistance auditable by joining canonical document artifacts, globally scoped evidence identity, local generation, claim-level support records, deterministic repair, and explicit selective failure in one trace. The implementation establishes these software contracts; it does not by itself establish that the verifier is calibrated, that repair improves human-judged support, or that the workflow benefits researchers. The release and manuscript therefore fail closed until those empirical and governance gates are supported by real artifacts.

Limitations

The current support scorer is an uncalibrated lexical heuristic and may miss paraphrases, numerical contradictions, negation, or evidence distributed across modalities. Deterministic deletion-based repair can discard useful nuance and may abstain on answerable questions; second-pass agreement with the same scorer is not independent validation. The system depends on PDF extraction quality, and scanned pages, equations, tables, figures, captions, and multi-column reading order can fail differently. Model acquisition, document acquisition, and package installation may require external network access even though strict-local analysis rejects public endpoints. Local-model capability, latency, memory, and throughput do not generalize beyond measured model builds, quantizations, prompts, and hardware. The intended user population and workflow benefit require a real, ethically reviewed pilot. The planned held-out set may cover only a limited set of scientific domains, document languages, and publication formats. ScholAR is not suitable for autonomous high-stakes scientific, clinical, legal, or safety decisions; users must inspect source evidence and primary papers.

Ethical Considerations

Local processing can reduce document disclosure, but it does not eliminate privacy risk: PDFs, extracted text, figures, embeddings, traces, prompts, and model outputs remain on the device and require deletion, access-control, and retention practices appropriate to the user's environment. The interface discloses the active network mode, and the paper distinguishes external acquisition from strict-local analysis. Benchmark documents and model artifacts must be used under their licenses. Human claim annotation and the researcher pilot must not begin before the responsible institution provides the applicable ethics or IRB determination; consent, compensation, de-identification, exclusions, and data retention must then follow the approved process. Citation-like output may create unwarranted trust, so the system exposes source evidence and selective failure and must not market a retrieved page as proof that a statement is true.

References

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A Release Artifact Contract

The evaluation release package contains: (1) a frozen expected-key registry `expected_keys.json`; (2) raw execution JSONL traces and post-scoring JSONL evaluations; (3) hierarchically aggregated results in `aggregate.json`; (4) auto-rendered LaTeX tables carrying SHA-256 provenance binds; (5) an execution manifest logging code commit, invocation flags, dataset identity, model hashes, quantization settings, prompt hashes, random seeds, hardware specs, timestamp, and status counts; and (6) a complete SHA-256 artifact manifest. The validator enforces strict key equality, rejects duplicate or truncated rows, verifies denominator preservation, checks table checksums against aggregate summaries, and audits for anonymity compliance.

B Answer Trace Fields

Every query processed by `AnswerPipelineService` produces a structured `AnswerTrace` recording:

- **Provenance:** Request ID, document ID, session ID, user question, random seed, model tag, prompt hash, and Git commit SHA.
- **Retrieval Telemetry:** Per-chunk candidate scores and ranks across lexical (BM25), dense text, crop CLIP, full-page ColQwen2, and modality channels; RRF fused ranks; sufficiency decisions and reason codes.
- **Prompts & Completions:** Exact prompt text, evidence block content hashes, raw model completion, normalized text, and final post-repair text.
- **Citation Metadata:** Source-scoped evidence identities, resolved document pages and bounding coordinates, and citation origins (ModelEmitted, ApplicationImputed, or Remapped).
- **Verification & Repair:** Pre- and post-repair atomic claims with character spans $[s_c, e_c)$, lexical overlap scores, numerical contradiction flags, support labels, concrete text diffs, and abstention codes.
- **System Metrics:** Stage-by-stage execution latencies (retrieval, generation, verification, repair), token throughput, and hardware tier descriptor.

C Human-Evidence Safety

Template directories for claim calibration, judge validation, researcher pilots, and institutional ethics determinations begin with an explicit no-data banner. Machine envelopes enforce strict metadata guards:

- `artifact_type`: `"template"`
- `study_status`: `"NOT_STARTED"`
- `contains_completed_data`: `false`
- `records`: empty list `[]`

Continuous integration automatically fails if any template directory contains populated participant or outcome records, ensuring unreviewed or nonexistent human empirical data cannot be merged or reported.